How replicable is intergroup-contact research?

A multi-method replicability and selective-reporting audit of 21 contact meta-analyses

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1 What this asks

Whether intergroup contact reduces prejudice is among the most studied claims in social psychology, and multiple meta-analyses have aggregated the evidence, generally with positive results. A mean effect, on its own, however, cannot say whether the significant findings behind it would hold up on repetition, nor how much of the published record was filtered to produce them. We therefore ask three questions of each body of contact research:

1. **How replicable are the significant findings?** If the significant contact findings in a literature were rerun at their original design and sample size, what share would again be significant? This is *unconditional mean power* (Brunner & Schimmack, 2020), estimated primarily by z-curve 2.0 and cross-checked with p-curve and a maximum-likelihood power estimator (Section 3).
2. **How selectively reported is the record?** How much of the significant record has been *selected for significance* (through publication bias, analytic flexibility, or other questionable research practices)? This is estimated with a Test of Excess Statistical Significance (Askarov et al., 2023; Stanley et al., 2021) and a three-parameter selection model (3PSM) (Hedges & Vevea, 2005) (Section 4).
3. **How big is the effect once selection is corrected?** Where part of a record was filtered, its naive meta-analytic mean is inflated; three bias-adjusted estimators ask what average effect survives the correction (Section 5).

We answer them with the full replicability-and-selective-reporting battery from a Stage-1 in-principle-accepted registered report (Wallrich et al., in press). The value of the battery is triangulation: methods that rest on different assumptions either converge on the same verdict for a literature or expose where a single-method reading would have misled.

! In brief

Pooled across **21 contact meta-analyses** (1,891 independent samples in the pooled prejudice/attitude curve once overlapping studies are deduplicated), replicability falls monotonically with experimental control: the estimated replication rate of significant prejudice/attitude effects is **0.85 for cross-sectional surveys, 0.70 for quasi-experiments, 0.63 for field experiments and 0.42 for lab experiments**. Contact *type* barely orders replicability (the types that sit low are those studied almost entirely in brief lab paradigms), and the same ordering recurs when the 21 meta-analyses are audited one at a time.

Selective reporting is a separate axis: excess significance concentrates in the experimental and intervention paradigms (imagined, modern lab/field, digital and mediated contact, most extremely Hsieh's field experiments), while the cleanest records are those deliberately assembled to include unpublished or preregistered work, however well or badly they replicate.

The corrected effect sizes complete the picture: the mean effects of the accumulated-contact, extended-contact and applied-intervention literatures survive every bias adjustment essentially unchanged, whereas the brief experimental paradigms shrink towards zero. Where selection is suppressed by design, as in Lowe's preregistered experiments, nothing flags and the average experimental contact effect is simply small (a standardised effect of $d \approx 0.1$ at most under every estimator).

Provisional: based on a limited number of bootstrap iterations

The registered report runs every method at 5,000 resamples and 5,000 bootstraps. This draft is a development run at reduced counts (300/300 for z-curve, p-curve and ESS; 250/250 for the 3PSM and bias-adjusted estimates; ML power at 20 resamples without a bootstrap CI). This adds Monte-Carlo noise and widens the CIs; every number is provisional pending the final 5,000/5,000 run. Details in Section 8.4.

2 The evidence base: 21 meta-analyses

We assembled a **pooled effect-level database** of **21 contact meta-analyses** (every one we could locate that reports per-study data), oriented so that positive effects indicate prejudice *reduction*. Data extraction for every source was validated by reproducing its published pooled estimate. Meta-analyses that report only pooled estimates, with no per-study data, cannot be audited and are listed in Section 12.

Meta-analysis	Contact studied	Samples	Effects
Pettigrew & Tropp (2006)	Foundational survey/quasi/experimental record	714	714
Paolini et al. (2024)	Positive vs. negative contact (valenced)	236	236
Paluck et al. (2021)	Modern contact field and lab experiments	155	508
Reimer & Sengupta (2023)	Contact & collective action (disadvantaged groups)	114	386
Zhou et al. (2019)	Extended contact + cross-group friendship	95	347
Maunder & White (2019)	Contact interventions for mental-health stigma	109	295
da Costa et al. (2024)	Digital / online contact	91	289
Lemmer & Wagner (2015)	Applied field contact interventions	123	123
Miles & Crisp (2014)	Imagined contact (brief experiments)	71	125
Van Assche et al. (2023)	Contact under threat (archival surveys)	66	225
Smith et al. (2009)	Contact & sexual prejudice	61	83
Banas et al. (2020)	Mediated (parasocial/vicarious) contact	78	78
Burnes et al. (2019)	Ageism interventions (contact arms)	57	70
McManus et al. (2021)	Attitudes to intellectual disability	54	131
Apriceno & Levy (2023)	Ageism programs (no per-study N)	125	173
Hsieh et al. (2022)	Real-world prejudice-reduction field experiments	39	53
Lowe (2025)	Preregistered contact experiments (benchmark)	41	41
Rahmani et al. (2025)	Narrative persuasion (intergroup subset)	22	128
Tondok et al. (2024)	Cooperative learning field experiments	18	18
Armstrong et al. (2017)	Disability contact interventions (children)	9	9
Griffiths et al. (2014)	Mental-illness stigma RCTs (contact arms)	5	5

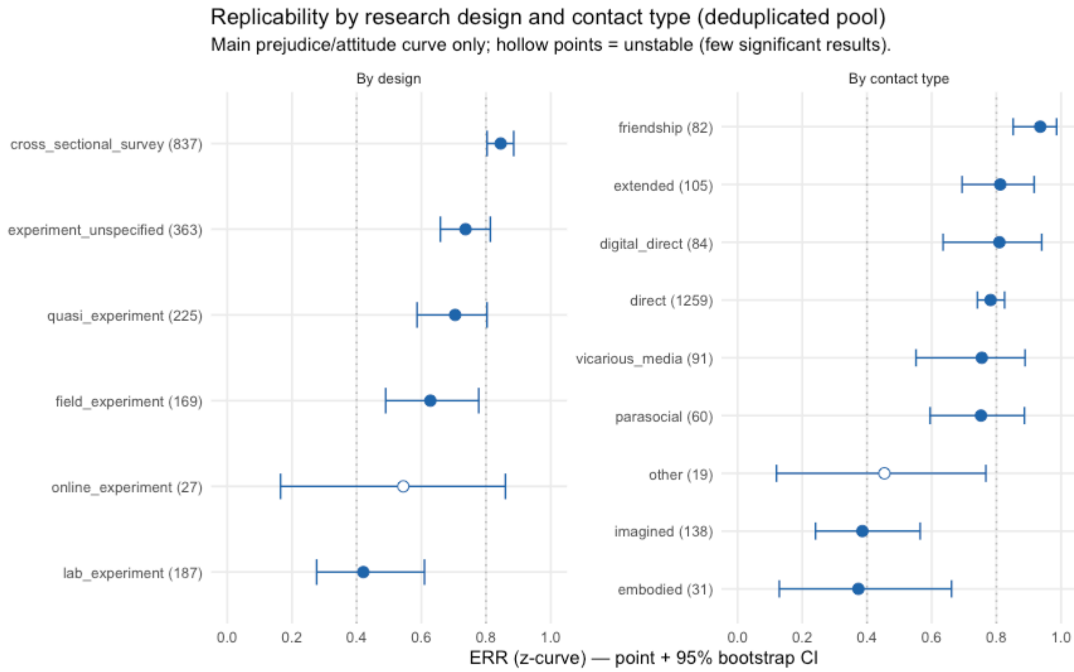
Note. Two sets of outcomes are excluded from the overall tests of contact–prejudice associations: Reimer & Sengupta's collective-action intentions, which measure a different outcome from prejudice (they are audited as their own separate curve in Section 3), and the factualknowledge-about-ageing measures inside Burnes and Apriceno (only their attitude effects are pooled).

The answers to all three questions for every literature are tabulated in Section 9; the sections that follow present question-specific displays. The three questions are taken in turn (Section 3, Section 4, Section 5), and Section 6 applies all three in depth to four contrasting literatures. Throughout, the field-wide comparisons use each meta-analysis's full contact sample; the case studies and subgroup contrasts use narrower, prespecified focal constructions built on each source's own coding, and Section 10 reconciles the two where they differ.

3 Question 1: How replicable are the significant findings?

The headline replicability metric is z-curve's **Expected Replication Rate (ERR)**: the estimated share of a literature's significant findings that would be significant again if rerun at the original design and sample size. It depends on statistical power, on selection for significance and on the base rate of true hypotheses, and it is estimated from the significant test statistics alone, so it can be computed for any literature regardless of how its record was filtered (see Section 8 for the estimator; ~0.80 and ~0.40 serve as rough anchors for well-powered and fragile literatures respectively).

Pooling the 21 meta-analyses and **deduplicating** overlapping samples turns *source meta-analysis* from the unit of analysis into a provenance tag, and lets the battery be cut by the variables that should actually drive replicability (research **design** and **contact type**), no longer confounded with which meta a study came from. The main curve covers the prejudice/attitude outcomes, including Van Assche's archival contact–prejudice surveys and the mental-illness-stigma interventions, under the scope set in Section 2. (This pooled analysis asks the reader to trust the deduplication and orientation choices before the full methods; Section 3.3 shows the same pattern with neither.)



Stratum	Level	k	ODR	ERR	EDR	PSSS
Design	cross_sectional_survey	837	0.72 [0.69, 0.76]	0.85 [0.80, 0.89]	0.48 [0.29, 0.80]	0.17 [0.06, 0.26]
Design	experiment_unspecified	363	0.56 [0.51, 0.61]	0.74 [0.66, 0.81]	0.41 [0.18, 0.72]	0.04 [−0.16, 0.18]
Design	field_experiment	169	0.39 [0.33, 0.47]	0.63 [0.49, 0.78]	0.15 [0.10, 0.59]	0.13 [0.02, 0.23]
Design	lab_experiment	187	0.47 [0.39, 0.55]	0.42 [0.28, 0.61]	0.20 [0.08, 0.47]	0.18 [0.05, 0.29]
Design	longitudinal_survey	15	0.87 [0.67, 1.00]	—	—	0.10 [−0.21, 0.38]
Design	online_experiment	27	0.25 [0.11, 0.44]	0.54 [0.16, 0.86]	0.11 [0.06, 0.74]	−0.26 [−0.65, 0.13]
Design	quasi_experiment	225	0.48 [0.42, 0.55]	0.70 [0.59, 0.80]	0.58 [0.22, 0.74]	0.00 [−0.12, 0.09]
Contact type	digital_direct	84	0.56 [0.45, 0.68]	0.81 [0.64, 0.94]	0.58 [0.14, 0.93]	0.14 [−0.04, 0.28]
Contact type	direct	1259	0.61 [0.58, 0.63]	0.78 [0.74, 0.83]	0.43 [0.24, 0.73]	0.17 [0.10, 0.22]

Stratum	Level	k	ODR	ERR	EDR	PSSS
Contact type	embodied	31	0.53 [0.35, 0.71]	0.37 [0.13, 0.66]	0.19 [0.06, 0.58]	0.14 [−0.04, 0.36]
Contact type	extended	105	0.76 [0.67, 0.84]	0.81 [0.69, 0.92]	0.61 [0.21, 0.89]	0.30 [0.04, 0.50]
Contact type	friendship	82	0.78 [0.68, 0.87]	0.94 [0.85, 0.99]	0.91 [0.40, 0.98]	−0.06 [−0.62, 0.31]
Contact type	imagined	138	0.45 [0.37, 0.55]	0.39 [0.24, 0.56]	0.20 [0.07, 0.46]	0.16 [0.05, 0.27]
Contact type	other	19	0.42 [0.18, 0.68]	0.45 [0.12, 0.77]	0.11 [0.06, 0.57]	0.24 [−0.03, 0.56]
Contact type	parasocial	60	0.79 [0.68, 0.88]	0.75 [0.60, 0.89]	0.49 [0.14, 0.84]	0.07 [−0.07, 0.22]
Contact type	vicarious_media	91	0.47 [0.38, 0.58]	0.75 [0.55, 0.89]	0.69 [0.20, 0.88]	0.21 [0.02, 0.37]
Separate curve	Collective action (Reimer)	114	0.46 [0.36, 0.55]	0.69 [0.53, 0.85]	0.41 [0.11, 0.77]	—

Pooled and deduplicated, the primary prejudice/attitude literature has an overall ERR of 0.76 [0.73, 0.80] across 1,891 independent samples, with a modest but non-zero selection signal (PSSS 0.14 [0.08, 0.20]). The interest is in how that average splits by design; the callout below carries the numbers. The gradient is read over the four design categories whose degree of experimental control is identifiable, which cover 1,418 of the 1,891 samples. The remaining strata appear in the figure but are not placed on the gradient: *experiment_unspecified* (363 samples, ERR 0.74) pools studies whose design could not be classified from the source coding, and the online-experiment and longitudinal-survey strata are too small for a stable fit (hollow points). The survey stratum includes Van Assche's very large archival contact surveys, assembled from existing datasets without regard to significance; their power reinforces its position at the top of the gradient, though Section 4 notes a narrow excess-significance flag even for that record.

> ! The headline finding
>
> Estimated replication rates of significant prejudice/attitude effects, on the deduplicated pool: **surveys 0.85 → quasi-experiments 0.70 → field experiments 0.63 → lab experiments 0.42**, a clean monotonic gradient over the four classifiable designs. The survey-vs-lab contrast is 0.41 [0.25, 0.58], and the survey-vs-quasi and quasi-vs-lab contrasts also exclude zero; the field-vs-lab estimate points the same way but its interval narrowly includes zero (0.20 [−0.00, 0.40]). This is a gradient in *replicability*, not a ranking of causal validity: the well-identified designs sit at the fragile end precisely because they are hard and expensive to power.

3.1 Design or contact type?

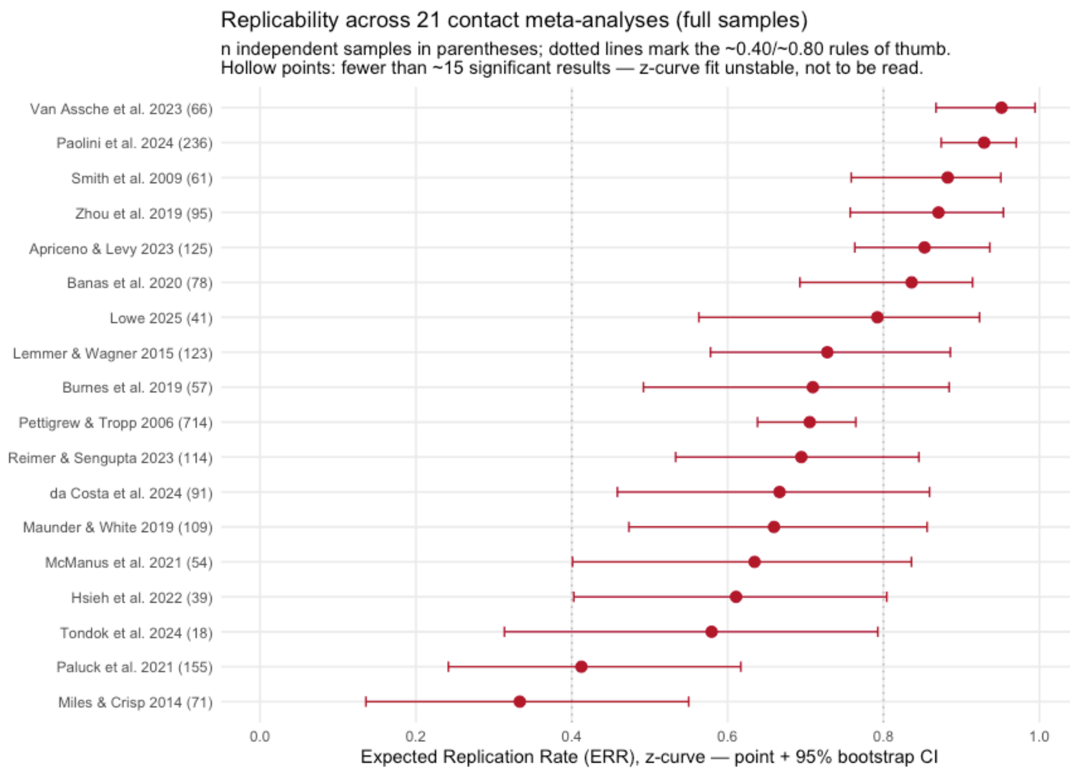
Contact *type* mostly does not order replicability. Cross-group friendship (ERR 0.94), extended, digital, vicarious-media, direct and parasocial contact all sit near the top of the figure; the two low types are imagined contact (0.39) and embodied/VR contact (0.37). But the extremes track design composition: friendship and extended contact are overwhelmingly survey measures, while imagined and embodied contact are overwhelmingly lab manipulations. The directness of contact, the axis the theoretical literature has emphasised, does not predict which effects replicate: extended and vicarious contact are as indirect as imagined contact and replicate as well as direct contact does. These are separate marginal cuts of the same pool, so they cannot establish that contact type adds nothing once design is held fixed; what they show is that design orders replicability strongly and cleanly, while contact type does not order it on its own.

3.2 Collective action, and the valence split

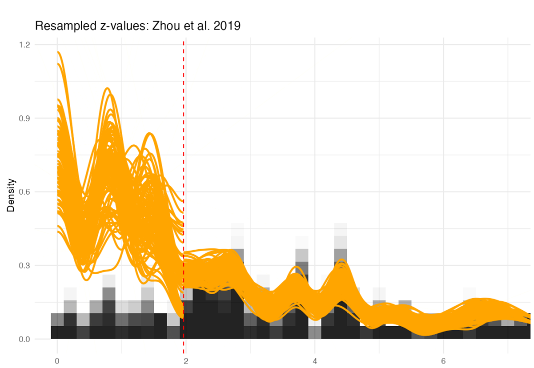
Collective action, audited as its own curve because it is a different outcome from prejudice, sits in the mixed-experimental band (ERR 0.69), not with the surveys. Splitting the prejudice pool by contact valence, positive contact (ERR 0.80) and negative contact (0.94) replicate equally well: the newer negative-contact literature, which predicts *increased* prejudice, is not the less reliable of the two. (The computationally intensive 3PSM and ML-power estimators are not run on the large pooled curves; their per-literature results appear in the following sections.)

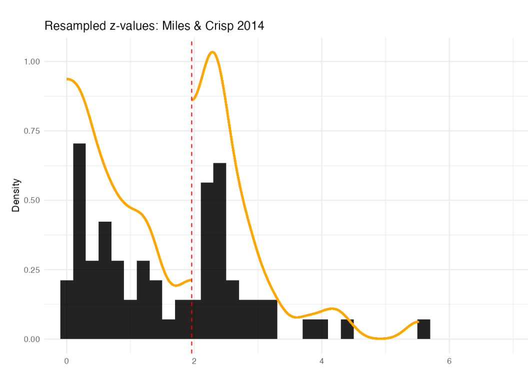
3.3 The pattern recurs, literature by literature

The pooled gradient could in principle be an artefact of the pooling itself: of the deduplication rules, the orientation choices, or a few very large sources dominating the strata. This check removes all of that: the battery is run on each meta-analysis's *full* contact sample at its own participant-pool unit, with **no cross-meta deduplication**, so every literature is audited exactly as its authors assembled it (the ERR column of the master table in Section 9).



The literature-by-literature ordering supports the pooled result. Each source here is a mixed literature rather than a design stratum, so this check cannot re-estimate the gradient; however, the survey- and archival-dominated literatures cluster at the top of the figure, together with two literatures assembled to resist selection (Lemmer & Wagner's applied interventions and Lowe's preregistered experiments), regardless of whether the contact they study is direct or indirect. The fragile end is again the brief experiments: Paluck's modern experiments (ERR 0.41) and imagined contact (0.33) are the two lowest stable estimates, with Hsieh's field experiments between them; the foundational Pettigrew & Tropp record sits at 0.71. Three meta-analyses (Griffiths, Armstrong, Rahmani) have too few significant results for a stable z-curve and are not interpreted; their rows are flagged.





Maunder & White's mental-illness-stigma interventions (0.66) sit in the mixed-experimental range alongside the other applied literatures. Maunder is off the design gradient by construction, not by omission: its source never coded per-study randomisation (the effects pool randomised, non-randomised and before–after designs), so its studies enter the pooled curve as *experiment_unspecified*.

3.4 Do other estimators agree?

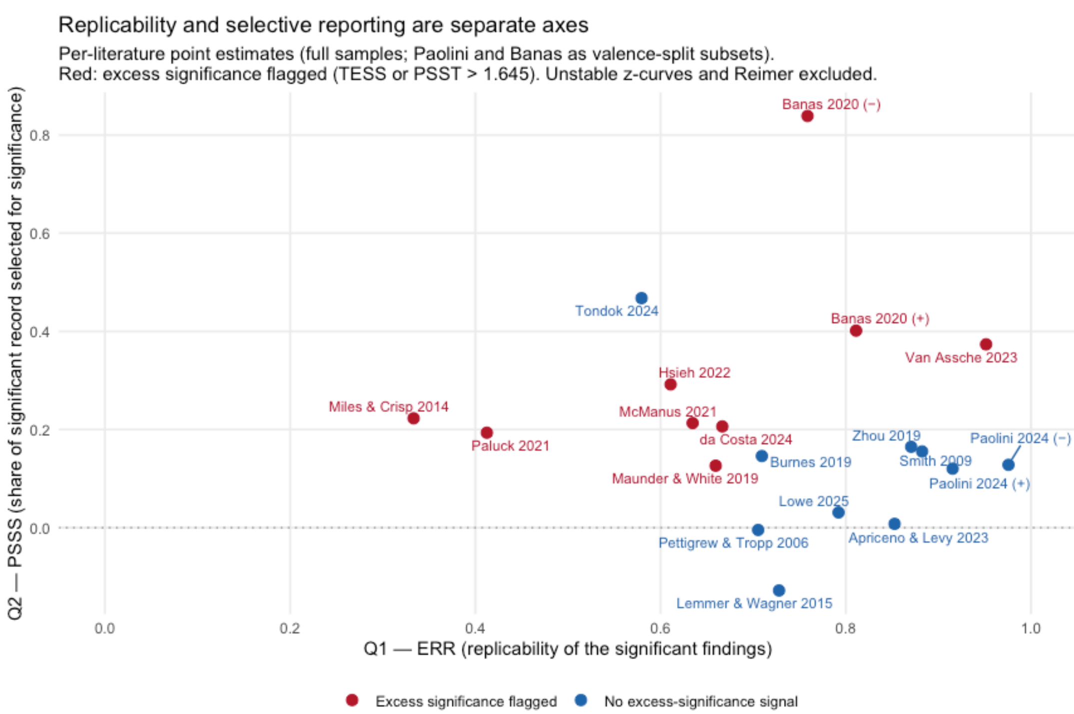
The gradient and the weak-end verdict lean on z-curve's ERR. Two cross-check estimators with different assumptions are therefore run on the same 21 full samples: p-curve and maximum-likelihood (ML) mean power. ML power needs per-study sample sizes, which Apriceno & Levy do not report, so it is omitted there; where the case studies quote the narrower focal constructions instead, they say so.

Meta-analysis	ERR	p-curve	ML power (pt)
Van Assche et al. 2023	0.95 [0.87, 0.99]	0.99 [0.99, 0.99]	0.95
Paolini et al. 2024	0.93 [0.87, 0.97]	0.99 [0.99, 0.99]	0.89
Smith et al. 2009	0.88 [0.76, 0.95]	0.99 [0.97, 0.99]	0.86
Zhou et al. 2019	0.87 [0.76, 0.95]	0.99 [0.99, 0.99]	0.88
Apriceno & Levy 2023	0.85 [0.76, 0.94]	0.99 [0.99, 0.99]	—
Banas et al. 2020	0.84 [0.69, 0.91]	0.98 [0.93, 0.99]	0.82
Lowe 2025	0.79 [0.56, 0.92]	0.97 [0.69, 0.99]	—
Lemmer & Wagner 2015	0.73 [0.58, 0.89]	0.98 [0.92, 0.99]	0.77
Burnes et al. 2019	0.71 [0.49, 0.88]	0.93 [0.74, 0.98]	0.69
Pettigrew & Tropp 2006	0.71 [0.64, 0.76]	0.98 [0.97, 0.99]	0.74
Reimer & Sengupta 2023	0.69 [0.53, 0.85]	0.98 [0.91, 0.99]	0.69
da Costa et al. 2024	0.67 [0.46, 0.86]	0.93 [0.78, 0.99]	0.65
Maunder & White 2019	0.66 [0.47, 0.86]	0.89 [0.72, 0.96]	0.59
McManus et al. 2021	0.63 [0.40, 0.84]	0.92 [0.68, 0.98]	0.64
Hsieh et al. 2022	0.61 [0.40, 0.80]	0.98 [0.87, 0.99]	0.78
Tondok et al. 2024	0.58 [0.31, 0.79]	0.99 [0.81, 0.99]	0.63
Paluck et al. 2021	0.41 [0.24, 0.62]	0.59 [0.34, 0.80]	0.41
Miles & Crisp 2014	0.33 [0.14, 0.55]	0.43 [0.13, 0.68]	0.35
Griffiths et al. 2014†	—	0.50 [0.05, 0.91]	0.51
Armstrong et al. 2017†	—	0.52 [0.09, 0.81]	0.33
Rahmani et al. 2025†	—	0.99 [0.69, 0.99]	0.78

Maximum-likelihood mean power tracks ERR across the stable literatures, with Hsieh the one clear divergence (0.78 vs 0.61); because ML power uses the sample sizes that z-curve ignores, this agreement shows the weak-end verdict does not depend on the z-curve model. p-curve separates the same weak literatures without reproducing the levels: biased upward under effect-size heterogeneity, it sits near the ceiling for every stable literature and collapses only at the weak end (0.59 Paluck, 0.43 Miles). What the estimators agree on is therefore that imagined contact and the modern experiments form the weak end, not the exact ordering or the absolute level; ML power also shares z-curve's methodological lineage rather than offering a fully independent one.

4 Question 2: How selectively reported is the record?

Replicability asks whether the *significant* findings would hold up; selective reporting asks how much of the *record* was filtered to produce them. The battery estimates both precisely because they can disagree; in the contact literature they do. The selective-reporting metrics work by comparing what a literature reports against what its power could produce: **PSSS** estimates the proportion of significant results that exist *because* they were significant (a conservative lower bound), the one-tailed **TESS/PSST** statistics test whether the excess of significant results is too large to dismiss (flagged above 1.645), and the **3PSM** estimates the relative publication odds of a non-significant versus a positive-significant result (Section 8.2 for full definitions).



The figure carries the section's central claim: the two axes are close to independent. Every combination occurs: highly replicable and clean (Pettigrew & Tropp, both of Paolini's valence subsets), highly replicable yet flagged (Van Assche, Banas's positive-contact subset), moderately replicable with the strongest selection signature in the set (Hsieh), and fragile with clear selection (Miles's imagined contact). Hsieh's field experiments pair a moderate ERR (0.61 on its full sample) with 76% of tests significant against an expected discovery rate of 0.13. On its narrower extended-contact-only focal construction, Zhou pairs the highest replicability (ERR 0.82) with the largest PSSS (0.38 [0.02, 0.62]): on its full sample the excess-significance signal does not reach significance (PSSS 0.16 [−0.28, 0.48]), but on the focal construction both test statistics clear the threshold (PSST 2.38, TESS 2.73); the PSSS interval is wide and only just excludes zero, but it registers selection by the pre-specified rule in a literature that carries near-top replicability. A single replicability index would miss all of this: it takes the excess-significance and selection-model methods to show that even well-replicating corners of the contact literature sit on a filtered evidence base.

Where, then, is the record most filtered? The strongest excess-significance signals cluster in the experimental and intervention paradigms — imagined contact, Paluck's modern experiments, Hsieh's field experiments, digital contact and mediated contact, with Hsieh the extreme (exact statistics in the table below). Weaker signals are not confined to the experiments, however: Van Assche's archival surveys, McManus's intellectual-disability studies and Armstrong's small nine-study record also cross the threshold, and Maunder flags on PSST. Statistics this close to the threshold are weaker evidence than the concentrated signals above, so the flag is best read as a matter of degree rather than a clean experimental/correlational divide. In the deduplicated pool the overall signal is modest but non-zero (PSSS 0.14 [0.08, 0.20]).

Meta-analysis	k	ODR	EDR	PSSS	TESS	PSST	3PSM ns: +sig
Paluck et al. 2021	155	0.47 [0.37, 0.56]	0.18 [0.07, 0.51]	0.19 [0.09, 0.31]	4.52 [0.65, 8.98]	3.41 [1.61, 5.52]	0.59 [0.22, 1.27]
da Costa et al. 2024	91	0.55 [0.44, 0.66]	0.41 [0.10, 0.80]	0.21 [0.05, 0.35]	4.22 [−0.69, 8.79]	3.08 [0.69, 5.39]	0.53 [0.19, 1.41]
Miles & Crisp 2014	71	0.47 [0.35, 0.59]	0.22 [0.07, 0.52]	0.22 [0.11, 0.36]	4.22 [1.23, 7.64]	2.97 [1.52, 4.48]	0.29 [0.03, 0.80]

Meta-analysis	k	ODR	EDR	PSSS	TESS	PSST	3PSM ns: +sig
Hsieh et al. 2022	39	0.76 [0.60, 0.90]	0.13 [0.09, 0.43]	0.29 [0.11, 0.48]	4.18 [0.64, 7.69]	2.60 [0.92, 4.35]	0.19 [0.03, 0.65]
Banas et al. 2020	78	0.81 [0.72, 0.90]	0.79 [0.27, 0.91]	0.18 [0.06, 0.33]	2.78 [−0.37, 6.52]	2.19 [0.73, 4.03]	0.30 [0.11, 0.79]
Armstrong et al. 2017†	9	0.56 [0.22, 0.89]	—	0.35 [−0.09, 0.78]	2.64 [−1.60, 5.68]	1.56 [−0.46, 2.86]	0.19 [0.03, 4.81]
McManus et al. 2021	54	0.59 [0.43, 0.74]	0.30 [0.10, 0.75]	0.21 [−0.12, 0.45]	2.40 [−3.72, 6.96]	1.80 [−0.89, 4.08]	0.68 [0.21, 2.51]
Van Assche et al. 2023	66	0.96 [0.89, 1.00]	0.78 [0.32, 0.99]	0.37 [−0.04, 0.68]	1.90 [−2.15, 5.50]	1.85 [−0.17, 3.44]	0.09 [0.00, 0.36]
Maunder & White 2019	109	0.43 [0.32, 0.52]	0.58 [0.17, 0.84]	0.13 [−0.07, 0.26]	1.64 [−4.17, 6.54]	1.87 [−0.80, 4.33]	1.50 [0.64, 3.19]
Burnes et al. 2019	57	0.55 [0.42, 0.70]	0.36 [0.11, 0.87]	0.15 [−0.01, 0.33]	1.33 [−1.86, 4.70]	1.37 [−0.06, 2.84]	0.38 [0.12, 1.01]
Tondok et al. 2024	18	0.89 [0.72, 1.00]	0.12 [0.07, 0.43]	0.47 [−0.41, 1.00]	0.93 [−2.36, 4.24]	1.02 [−0.84, 2.56]	0.20 [0.00, 2.37]
Griffiths et al. 2014†	5	0.40 [0.00, 0.80]	—	0.05 [−0.21, 0.55]	−0.19 [−1.83, 2.45]	0.14 [−0.66, 1.40]	0.58 [0.04, 98.80]
Smith et al. 2009	61	0.77 [0.66, 0.87]	0.73 [0.26, 0.94]	0.16 [−0.28, 0.43]	−0.25 [−3.94, 2.68]	0.75 [−1.19, 2.15]	0.97 [0.39, 2.36]
Zhou et al. 2019	95	0.79 [0.69, 0.87]	0.77 [0.27, 0.95]	0.16 [−0.28, 0.48]	−0.28 [−4.89, 4.67]	0.96 [−1.42, 3.18]	0.87 [0.29, 2.08]

Meta-analysis	k	ODR	EDR	PSSS	TESS	PSST	3PSM ns: +sig
Lowe 2025	41	0.39 [0.27, 0.54]	0.28 [0.12, 0.92]	0.03 [−0.18, 0.22]	−0.87 [−4.27, 3.00]	0.27 [−1.23, 2.21]	1.79 [0.54, 5.30]
Apriceno & Levy 2023	125	0.74 [0.66, 0.81]	0.34 [0.20, 0.91]	0.01 [−0.67, 0.28]	−2.29 [−11.72, 4.88]	0.11 [−4.47, 3.25]	0.44 [0.23, 0.91]
Paolini et al. 2024	236	0.86 [0.81, 0.90]	0.81 [0.38, 0.97]	0.03 [−0.18, 0.21]	−2.44 [−9.79, 6.05]	0.48 [−2.75, 4.46]	0.27 [0.16, 0.39]
Rahmani et al. 2025†	22	0.32 [0.09, 0.55]	—	−0.13 [−0.47, 0.16]	−2.69 [−6.46, 1.19]	−0.72 [−2.42, 1.01]	5.54 [0.37, 99.20]
Lemmer & Wagner 2015	123	0.36 [0.28, 0.44]	0.39 [0.13, 0.82]	−0.13 [−0.26, −0.02]	−6.25 [−9.40, −3.26]	−1.63 [−3.00, −0.33]	1.96 [1.23, 3.49]
Pettigrew & Tropp 2006	714	0.56 [0.52, 0.59]	0.39 [0.17, 0.66]	−0.00 [−0.09, 0.11]	−6.39 [−10.92, 0.22]	−0.11 [−2.10, 2.77]	1.07 [0.76, 1.39]
Reimer & Sengupta 2023	114	0.46 [0.36, 0.55]	0.41 [0.11, 0.77]	—	—	—	—

Because Paolini's and Banas's records mix positively and negatively valenced contact (whose hypothesised directions oppose), their direction-dependent statistics come from valence-split subsets, each oriented to its own hypothesis (`replicability_engine/run_valence_splits.R`; the oriented means reproduce the published per-valence pooled estimates). The split sharpens both verdicts: Paolini is clean in both halves and both means survive correction (Section 5), while Banas's signal, diluted in the pooled mixture, strengthens on both sides (positive contact PSSS 0.40 [0.10, 0.63]; the small negative-contact subset 0.84 [0.31, 1.00]). Mediated contact thereby joins Hsieh as a literature that replicates reasonably well on a visibly filtered record.

The clean end is anchored by exactly the records built to include unpublished work. Lemmer & Wagner's PSSS is significantly *negative* (−0.13 [−0.26, −0.02]), and its 3PSM puts the publication odds of a non-significant result at roughly twice those of a positive-significant one, a pattern compatible with a literature built on dissertations and unpublished interventions, though not diagnostic of it. Pettigrew & Tropp's PSSS is essentially zero (−0.00 [−0.09, 0.11]) with 3PSM odds near parity (1.07), consistent with a meta-analysis that deliberately gathered unpublished work. Neither Paolini, Zhou, Smith, Apriceno nor Lowe shows a positive excess-significance signal on their full samples. Because PSSS is a conservative lower bound, a near-zero value means excess significance was not *detected*: it does not bound selection from above, and appreciable undetected selection remains compatible with it. The clean end is therefore "no selection detected", not proof of its absence.

5 Question 3: How big is the effect once selection is corrected?

If a literature has been selected for significance, its naive meta-analytic mean is inflated, so the third question is what average effect survives once selection is corrected: which contact conclusions stand, which weaken, and which disappear. Three adjustment strategies with different assumptions are compared against the naive random-effects estimate, each on the dataset's own effect-size scale: trim-and-fill (Duval & Tweedie, 2000), included deliberately because several of the original meta-analyses relied on it as their bias check; PET-PEESE (Stanley & Doucouliagos, 2014), a regression-based correction using the effect–SE relationship; and the 3PSM-adjusted mean, the mean effect implied by the selection model. Trim-and-fill detects funnel asymmetry rather than p-value selection, and here it duly fails as a bias check: it trims Pettigrew & Tropp, where nothing flags, more than it moves Hsieh, the strongest-selected record in the audit, so passing it is weak reassurance.

dataset	Scale	Random effects	Trim-and-fill	PET-PEESE	3PSM-adjusted
McManus et al. 2021	d	0.71 [0.50, 0.91]	0.60 [0.27, 0.85]	0.29 [−0.14, 0.75]	0.62 [0.23, 1.03]
Smith et al. 2009	d	0.56 [0.48, 0.65]	0.56 [0.47, 0.68]	0.49 [0.38, 0.62]	0.56 [0.46, 0.66]
Armstrong et al. 2017	d	0.44 [0.18, 0.75]	0.29 [0.15, 0.65]	−0.01 [−0.66, 0.22]	0.24 [0.12, 0.66]
Rahmani et al. 2025	d	0.42 [0.14, 0.84]	0.46 [0.14, 0.86]	0.42 [0.06, 0.84]	0.68 [0.12, 1.38]
Maunder & White 2019	d	0.41 [0.33, 0.53]	0.41 [0.23, 0.53]	0.06 [−0.16, 0.33]	0.45 [0.28, 0.65]
Paolini et al. 2024 — negative contact	Fisher z*	0.39 [0.35, 0.44]	0.33 [0.30, 0.42]	0.38 [0.31, 0.45]	0.36 [0.24, 0.43]

dataset	Scale	Random effects	Trim-and-fill	PET-PEESE	3PSM-adjusted
Paluck et al. 2021	d	0.36 [0.28, 0.43]	0.23 [0.16, 0.40]	−0.13 [−0.36, 0.07]	0.28 [0.16, 0.42]
Miles & Crisp 2014	d	0.36 [0.25, 0.45]	0.22 [0.15, 0.32]	−0.36 [−0.63, −0.13]	0.18 [0.01, 0.34]
Burnes et al. 2019	d	0.35 [0.18, 0.53]	0.30 [−0.02, 0.49]	0.41 [−0.22, 0.75]	−0.36 [−1.38, 0.47]
Griffiths et al. 2014	d	0.33 [0.12, 0.52]	0.33 [0.12, 0.57]	1.78 [−2.18, 8.35]	0.29 [0.07, 0.55]
Lemmer & Wagner 2015	g	0.33 [0.27, 0.38]	0.39 [0.29, 0.44]	0.35 [0.28, 0.42]	0.39 [0.31, 0.48]
Banas et al. 2020 — negative contact	Fisher z*	0.32 [0.25, 0.42]	0.32 [0.23, 0.42]	0.21 [0.06, 0.37]	−0.26 [−16.12, 0.29]
Tondok et al. 2024	g	0.32 [0.25, 0.39]	0.32 [0.24, 0.40]	0.28 [0.01, 0.39]	0.24 [−0.79, 0.39]
Zhou et al. 2019	Fisher z	0.27 [0.23, 0.31]	0.30 [0.23, 0.35]	0.30 [0.24, 0.35]	0.26 [0.20, 0.31]
Apriceno & Levy 2023	g	0.25 [0.13, 0.37]	0.25 [0.10, 0.55]	0.34 [0.20, 0.46]	−0.28 [−0.85, 0.14]
da Costa et al. 2024	g (reduction = +)	0.24 [0.13, 0.36]	0.09 [−0.04, 0.35]	0.07 [−0.15, 0.32]	0.12 [−0.27, 0.35]
Banas et al. 2020 — positive contact	Fisher z	0.23 [0.17, 0.28]	0.15 [0.12, 0.26]	0.07 [−0.05, 0.22]	0.18 [0.10, 0.27]
Paolini et al. 2024 — positive contact	Fisher z	0.23 [0.19, 0.28]	0.13 [0.10, 0.31]	0.23 [0.18, 0.30]	0.10 [−0.04, 0.22]
Hsieh et al. 2022	d	0.22 [−0.04, 0.46]	0.18 [−0.05, 0.46]	0.25 [−0.36, 0.72]	−0.17 [−0.62, 0.33]

dataset	Scale	Random effects	Trim-and-fill	PET-PEESE	3PSM-adjusted
Pettigrew & Tropp 2006	Fisher z	0.21 [0.20, 0.22]	0.15 [0.13, 0.22]	0.20 [0.18, 0.21]	0.22 [0.19, 0.23]
Van Assche et al. 2023	Fisher z	0.21 [0.14, 0.29]	0.17 [0.09, 0.27]	0.20 [0.04, 0.30]	0.04 [−0.15, 0.24]
Lowe 2025	std. effect	0.11 [0.06, 0.16]	0.11 [0.04, 0.19]	0.10 [−0.01, 0.16]	0.14 [0.05, 0.21]

Where no selection was detected, the conclusions survive, and some strengthen. The accumulated-contact, extended-contact and applied-intervention literatures change little under any correction: Pettigrew & Tropp's naive mean of 0.21 (Fisher z, matching the published pooled r = −.21) holds at 0.15–0.22 across the corrections, and Zhou, Smith and both of Paolini's valence halves behave the same way (table above). Lemmer & Wagner's adjustments go *up*, as expected for a literature whose excess-significance statistics are negative, and Lowe's preregistered estimate stays small under every method. The evidence the contact hypothesis chiefly rests on therefore does not depend on trusting the uncorrected means, nor on believing any particular bias model.

Where selection was flagged, the corrections bite, and one conclusion disappears. Miles & Crisp's naive d = 0.36 falls to 0.22 under trim-and-fill and 0.18 under the 3PSM, and Paluck and Banas's positive-contact subset shrink similarly (table above). The clearest reversal is digital contact: da Costa's 0.24 shrinks to estimates indistinguishable from zero under all three corrections, every CI spanning zero. PET-PEESE goes further for the two weakest literatures, flipping Miles and Paluck negative, better read as "no evidence of a non-zero mean" than as genuinely negative effects.

Where selection is extreme or the record small, no single corrected value can be trusted. The 3PSM-adjusted mean turns erratic wherever its one-sided fallback meets a small or approximate record, flipping Hsieh, Burnes, Apriceno and Banas's 26-sample negative-contact subset to implausible values (table above). The mapping from diagnostics to adjustments is not one-to-one either: Zhou's focal set flags selection yet its mean barely moves, while Hsieh's corrections point in opposite directions. For those sources the defensible conclusion is uncertainty, not any single corrected estimate.

6 Four case studies: the three questions in action

The three questions so far describe the whole field; the case studies show what their combinations look like from inside a single literature. The four occupy distinct positions in the replicability × selection space of Section 4: **Pettigrew & Tropp** is replicable and clean (and contains the design gradient *within* a single meta-analysis), **imagined contact** is fragile and selected, **Hsieh's field experiments** are moderately replicable but extremely selected, and **Lowe's preregistered experiments** suppress selection by design. Each case answers the three questions in turn, from the battery run on its focal construction (Section 10).

6.1 Pettigrew & Tropp (2006): the foundational record, and the gradient in miniature

On replicability, the design gradient runs *within* the meta-analysis the contact hypothesis rests on: survey and field studies of accumulated contact (ERR 0.76) are more replicable than quasi-experiments (0.60), a difference of 0.15 [0.03, 0.30] that excludes zero, with pure experiments falling further to 0.51 (too few, $n = 36$, to separate cleanly). Because this contrast compares designs inside one meta-analysis (same inclusion rules, same coding, same era), it rules out the source-level differences that could confound the pooled gradient. Design remains entangled with setting, sample size and contact paradigm even within the record, but of the evidence here it comes closest to isolating design itself.

On selective reporting, the record shows no sign of selection on any indicator: PSSS −0.00 [−0.10, 0.09], and the 3PSM puts non-significant and positive-significant results at almost equal publication odds (1.07 [0.81, 1.37]), consistent with its deliberate inclusion of unpublished studies. Its published and unpublished effects do not differ in replicability (ERR 0.69 vs 0.75) and both show essentially zero selection (PSSS −0.00 vs −0.06), so there is no sign that the published record is inflated relative to the file drawer. The corrected effect follows suit: the naive mean (Fisher z 0.21, matching the published pooled r = −.21) holds at 0.20–0.22 under PET-PEESE and the 3PSM.

6.2 Imagined contact (Miles & Crisp 2014): the fragile extreme

Imagined contact has the lowest stable estimated replicability in the audit (ERR 0.28 [0.13, 0.51] on its 71-experiment focal construction), in the range of literatures that fared badly in large replication projects and consistent with its small two-cell designs. Every estimator agrees: p-curve (0.44) and ML power (0.34) place it just as low, so the weakness is genuine underpowering, not an artefact of z-curve's model. It also carries a clear selection signature (PSSS 0.22 [0.11, 0.33], TESS 4.0), and its bias-adjusted mean shrinks accordingly: the naive d = 0.34 falls to 0.21 under trim-and-fill and 0.18 under the 3PSM, with PET-PEESE flipping negative (−0.30), a known behaviour of that method under heterogeneity and small k that is better read as "no evidence of a non-zero mean" than as a genuinely negative effect.

The theoretical point comes from the comparison with the other indirect forms: extended and vicarious contact are just as indirect and replicate as well as direct contact, so the marginal pattern implicates the brief lab paradigm far more strongly than indirectness, though, as with the pooled cut, the two are not experimentally separable here.

6.3 Hsieh et al. (2022): the selection extreme

Hsieh's real-world field experiments show the strongest selection signature in the audit, and they demonstrate why replicability and selection cannot be collapsed into one index. On its focal construction (40 studies, 55 arms), their replicability is middling (ERR 0.62 [0.44, 0.81]), yet 77% of the resampled tests come out significant (ODR) against an expected discovery rate of only 0.13, far more "hits" than the studies' power can support, and the 3PSM estimates the publication odds of a non-significant result at just 0.17 times those of a positive-significant one, the most extreme weighting in the set.

The case also illustrates when bias correction stops working: with ~40 heterogeneous arms and selection this strong, the adjusted means are erratic (random effects 0.06 with a CI from −0.34 to 0.39; 3PSM-adjusted −0.82) and none should be taken at face value. A record this filtered can be *diagnosed*, but its true mean effect cannot be recovered from it.

6.4 Lowe (2025): the selection-resistant benchmark

Lowe's meta-analysis of preregistered contact experiments is the natural control on everything above, because the two features that make it selection-resistant are exactly the ones the battery is built to detect the absence of. Its studies are drawn from the AEA, EGAP and OSF registries, so registered nulls are retrieved whether or not they were published; and each effect is read off a preregistered primary outcome, which blocks within-study outcome selection. These features suppress the two main routes to selection rather than leaving them to be estimated away, though they cannot guarantee complete registration, adherence and retrieval.

> **!** Read Lowe's evidence directly, not through z-curve
>
> On the 25 clean experiments of the case-study focal set, a random draw of one effect per experiment leaves only ~7 significant tests, so the z-curve fit is unstable and its ERR should not be read (the bootstrap CIs are flagged; on the full 41-effect sample it is estimable at 0.79). The scarcity of significant results is consistent with the low discovery rate and small average effect: it makes z-curve uninformative, but does not by itself demonstrate suppressed selection. The case for selection resistance rests on the design (registry-based retrieval, preregistered outcomes) together with the absence of any positive selection signal; the informative evidence is direct: a low discovery rate and a small mean effect.

The rest of the battery, which runs in full on Lowe's effect-level data, confirms the direct reading: no selection indicator reaches significance (PSSS 0.12 [−0.06, 0.28], PSST 1.54), and the mean effect sits at 0.03–0.10 under the naive and every bias-adjusted estimator alike, with all CIs spanning zero. This is the mirror image of imagined contact: where selection is possible, the significant effects are unreliable; where it is suppressed by design, the average experimental contact effect is simply small.

7 What it adds up to

Replicability in intergroup-contact research is ordered by research design. On the deduplicated pool of 21 meta-analyses, the estimated replication rate of significant prejudice/attitude effects falls monotonically with experimental control, halving from the cross-sectional surveys (0.85) to the lab experiments (0.42). What makes the gradient credible is its recurrence: it is echoed when the 21 literatures are audited one at a time, it holds within the foundational Pettigrew & Tropp record itself, and the verdict on the weak end does not depend on the z-curve model, since p-curve and ML power also place imagined contact and the modern experiments lowest. Contact *type* does not order replicability on its own (extended and vicarious contact are as indirect as imagined contact and replicate as well as direct contact), though the marginal cuts here cannot fully separate type from the designs each type is studied with.

Selective reporting is a second, separate story. The strongest excess significance sits in the experimental and intervention paradigms (most extremely in Hsieh's field experiments, which report significance far above what their power can support), with weaker signals crossing the threshold in a few correlational and applied records too, and it can coexist with high replicability, as extended contact shows. The clean end is anchored by the literatures assembled to include unpublished work. A replicable literature, in other words, is not automatically an unfiltered one; the two verdicts have to be read together.

Correcting for that filtering translates the pattern into effect sizes. The core evidence for intergroup contact survives: accumulated real-world contact measured in surveys and field studies, extended contact and applied field interventions have estimated replication rates in the range where significant findings are expected to hold up, and their mean effects survive every bias adjustment essentially unchanged. Digital contact is the instructive counterexample: it replicates moderately (ERR 0.67), but it carries a clear excess-significance signal and its adjusted mean shrinks to values indistinguishable from zero (Section 5). The fragility is localised and predictable: it lives in the brief experimental paradigms, above all imagined contact (ERR ~0.3, with a clear selection signature and adjusted means one-third to one-half smaller), with the broader modern experiments not far behind. And where selection is suppressed by design, in Lowe's preregistered experiments, no indicator flags and the average experimental contact effect is simply small (a standardised effect of around 0.1 under every estimator; Section 6.4). On this evidence the contact hypothesis is carried by the observational and applied literatures (accumulated survey contact, extended contact, real-world interventions) more than by the brief experiments that are easiest to run and easiest to over-read.

! Replicable does not mean causal

The gradient runs the wrong way for causal inference: the designs that replicate best identify contact effects worst. High survey-literature ERR partly reflects the reproducibility of stable individual-differences correlations and shared-method variance in large samples; it speaks to the replicability of the *evidence*, not to causal identification. The audit's conclusion is about which bodies of evidence would hold up on repetition, not that surveys establish, better than experiments do, that contact causes prejudice reduction.

8 The battery: methods in full

The engine (`replicability_engine/engine.R`) runs the registered report's own analysis code (Wallrich et al., in press), wrapped in the same resample-then-bootstrap dependence handling. The *approach* — the choice of methods and how they are combined — was prespecified and accepted at Stage-1 review for IWO psychology. That acceptance covers the battery itself; its application to contact data here, including the adaptations in Section 8.4 and the bias-adjusted estimators, is our own and is not underwritten by the Stage-1 decision.

8.1 Replicability: mean power

z-curve 2.0 (Bartoš & Schimmack, 2022) fits a finite mixture of non-central distributions to the *significant* z-values (a selected literature cannot be fit on its full set of results), then extrapolates to the non-significant region. It yields:

- **ERR — Expected Replication Rate:** the mean power of the *significant* studies; the headline replicability index, and the quantity z-curve estimates most precisely.
- **EDR — Expected Discovery Rate:** the mean power of *all* studies (significant or not). It is estimated by extrapolation and is therefore far noisier than ERR.
- **ODR — Observed Discovery Rate:** the share of *reported* tests that are significant. A large gap of ODR above EDR is the signature of selection.
- **ARP — Actual Replication Prediction:** the mean of ERR and EDR, which bracket true replicability from above and below. Aggregated across six earlier large replication projects it landed within two points of the realised replication rate (Schimmack, 2022; Sotola, 2023). Our own check against the DARPA SCORE program's replications (`zcurve_predictive_accuracy/`; 257 significant originals from the official data release) is less encouraging: ARP = .78 against a realised rate of .55, an overestimate of about 23 points that is only partly attributable to SCORE's stricter success criterion. We therefore read ARP here as an optimistic summary of the ERR–EDR bracket rather than a calibrated prediction of any specific replication effort.

p-curve (Simonsohn et al., 2014) is the longer-established relative of z-curve and returns an estimate of the mean power of the significant studies. It is biased upward under effect-size heterogeneity (Carter et al., 2019) and discriminates replication success less well than z-curve (Sotola, 2023), so it is reported as a robustness check, not a headline.

ML power (Brunner & Schimmack, 2020) estimates the same quantity as ERR — the mean power of the significant results — by maximum likelihood: the significant test statistics are modelled as non-central chi-square draws whose underlying effects follow a gamma distribution, and the likelihood is maximised conditional on significance. It shares z-curve's lineage, but differs in model family and estimation (a continuous parametric effect distribution fit by ML, versus z-curve's finite mixture fit by EM on z-values) and it uses the studies' sample sizes, which z-curve ignores. Agreement between the two therefore shows the results are not an artefact of z-curve's specific mixture model. The registered report uses ML power in exactly this way, as a cross-check on ERR.

8.2 Selective reporting

Excess Statistical Significance (ESS) (Stanley et al., 2021) estimates the *expected* share of significant results from an unrestricted-least-squares mean effect and its heterogeneity, and compares it to the observed share. Following Askarov et al. (2023) this is converted into the headline selective-reporting quantity:

- **PSSS — proportion of significant results selected for significance.** Because the reported mean effect is itself inflated by selection, the expected significant share is biased upward and PSSS is biased downward, making it a conservative lower bound on selective reporting.
- **PSST — proportion of statistical significance test** asks whether the *observed share* of significant results exceeds the share the studies' estimated power could produce. **TESS — test of excess statistical significance** asks whether the *excess share* of significant results (observed minus expected) exceeds a 5% tolerance margin, i.e. whether the excess is too large to dismiss as minor. Both are one-tailed z-tests, significant above 1.645; following the guidance of Stanley et al. (2021), selective reporting is inferred when either is significant.

3PSM — three-parameter selection model (Hedges & Vevea, 2005) directly estimates the relative publication odds of non-significant versus positive-significant results. It rests on stricter assumptions than ESS (selection independent of sample size; a parametric effect-size distribution), so it is reported as supplementary context. Its primary output here is the non-significant-vs-positive-significant odds; the surprising-negative weight is secondary.

8.3 Dependence handling and effect sizes

Resample-then-bootstrap. Every method assumes independent p-values, which is violated when several tests come from one sample. Following the registered report, the engine draws one focal test per independent sample in each iteration (drawing one effect size where a sample contributes several): the point estimate of any quantity is its mean over `n_res` such resamples, and its 95% CI is the 2.5/97.5 percentile over `n_boot` bootstrap resamples, each resampling studies with replacement and then again drawing one effect size per sample.

Effect sizes. ESS, the 3PSM and the bias-adjusted estimates run directly on each meta-analysis's own standardised effect sizes and standard errors: Fisher-transformed r with SE $1/\sqrt{(n-3)}$ for the correlation-based datasets, and d/g with the reported or design-derived SE for the SMD-based ones, oriented so that positive = the hypothesised prejudice reduction. These statistics are computed within a single dataset, so no cross-dataset standardisation is needed and each dataset stays on its native scale. (The registered report, whose inputs are hand-coded p-values from primary articles, reconstructs an effect size via p → t → r → Fisher's Z_r; with meta-analytic inputs that reconstruction is unnecessary, which also decouples these methods from sample-size availability.)

Subgroup differences. There is no closed-form test for differences in these estimates, so a difference between two subgroups is judged significant iff the percentile 95% CI of the bootstrapped difference excludes zero.

8.4 How this differs from the registered-report pipeline

The registered report's pipeline is applied with four adaptations that follow from using meta-analytic contact data rather than hand-coded primary studies: where the meta-analytic setting makes a step unnecessary, that step is dropped.

1. **Meta-analytic inputs.** The registered report hand-codes focal p-values from primary articles; here the focal tests are the effect sizes (r, d, g) and sample sizes reported in published contact meta-analyses, with p-values computed from them, and the effect-size-based methods run on those effect sizes directly (Section 8.3).
2. **Direction convention.** Effects are oriented so that positive = the hypothesised prejudice reduction (for da Costa 2024 this means flipping the sign its data file uses; see Section 10). Direction feeds only the selective-reporting statistics; ERR and the other power estimates use |z| and are direction-free.
3. **Iteration counts.** The registered report runs every method at 5,000 resamples and 5,000 bootstraps. *This draft is a development run* at reduced counts: 300/300 for z-curve, p-curve and ESS; 250/250 for the 3PSM and the bias-adjusted estimates; ML power at 20 resamples with no bootstrap CI (with-replacement resamples occasionally stall its nested numerical integration inside compiled code, where a timeout cannot interrupt it). Reducing the counts adds Monte-Carlo noise to the point estimates and widens the percentile CIs, with the headline ERR and PSSS held to higher precision than the supplementary methods (see the engine README). Every result here is therefore provisional pending the final run at the registered report's 5,000/5,000 throughout, with the ML bootstrap either restored

under a worker-level guard or ML power reported alongside p-curve's bootstrap CI as the interval-bearing cross-check.

4. **Bias-adjusted estimates added.** Beyond the registered report's battery, each meta-analysis also gets bias-adjusted mean-effect estimates: random-effects, trim-and-fill, PET-PEESE and the 3PSM-adjusted mean (Section 5). These are exploratory additions beyond the registered report's pipeline.

9 Appendix: the three questions across the 21 meta-analyses

One row per meta-analysis, with the headline answer to each question, from the battery run on each meta-analysis's full contact sample.

The three questions across the 21 meta-analyses

Meta-analysis	k	Q1: ERR	Q2: excess significance	Q2: PSSS	Q3: naive mean	Q3: 3PSM-adjusted
Correlation-based sources (Fisher z)						
Van Assche et al. 2023	66	0.95 [0.87, 0.99]	flagged	0.37 [–0.04, 0.68]	0.21 [0.14, 0.29]	0.04 [–0.15, 0.24]
Paolini et al. 2024‡	236	0.93 [0.87, 0.97]	not flagged	+: 0.12 [–0.32, 0.44]; –: 0.13 [–0.82, 0.64]	+: 0.23 [0.19, 0.28]; –: 0.39 [0.35, 0.44]	+: 0.10 [–0.04, 0.22]; –: 0.36 [0.24, 0.43]

Point [95% bootstrap CI]; full samples, no cross-meta deduplication; ordered by ERR within group. Means are on each group's scale and cannot be compared across groups. Excess significance is flagged when TESS or PSST exceeds 1.645.

† Fewer than ~15 significant results: z-curve unstable, ERR not to be read.

‡ Valenced source: the Q2/Q3 cells show the valence-split values (+ positive, – negative contact, each oriented to its own hypothesis).

The three questions across the 21 meta-analyses

Meta-analysis	k	Q1: ERR	Q2: excess significance	Q2: PSSS	Q3: naive mean	Q3: 3PSM-adjusted
Zhou et al. 2019	95	0.87 [0.76, 0.95]	not flagged	0.16 [–0.28, 0.48]	0.27 [0.23, 0.31]	0.26 [0.20, 0.31]
Banas et al. 2020‡	78	0.84 [0.69, 0.91]	flagged	+: 0.40 [0.10, 0.63]; –: 0.84 [0.31, 1.00]	+: 0.23 [0.17, 0.28]; –: 0.32 [0.25, 0.42]	+: 0.18 [0.10, 0.27]; –: –0.26 [–16.12, 0.29]
Pettigrew & Tropp 2006	714	0.71 [0.64, 0.76]	not flagged	–0.00 [–0.09, 0.11]	0.21 [0.20, 0.22]	0.22 [0.19, 0.23]
SMD-based sources (d, g, or standardised effect)						
Smith et al. 2009	61	0.88 [0.76, 0.95]	not flagged	0.16 [–0.28, 0.43]	0.56 [0.48, 0.65]	0.56 [0.46, 0.66]
Apriceno & Levy 2023	125	0.85 [0.76, 0.94]	not flagged	0.01 [–0.67, 0.28]	0.25 [0.13, 0.37]	–0.28 [–0.85, 0.14]

Point [95% bootstrap CI]; full samples, no cross-meta deduplication; ordered by ERR within group. Means are on each group's scale and cannot be compared across groups. Excess significance is flagged when TESS or PSST exceeds 1.645.

† Fewer than ~15 significant results: z-curve unstable, ERR not to be read.

‡ Valenced source: the Q2/Q3 cells show the valence-split values (+ positive, – negative contact, each oriented to its own hypothesis).

The three questions across the 21 meta-analyses

Meta-analysis	k	Q1: ERR	Q2: excess significance	Q2: PSSS	Q3: naive mean	Q3: 3PSM-adjusted
Lowe 2025	41	0.79 [0.56, 0.92]	not flagged	0.03 [–0.18, 0.22]	0.11 [0.06, 0.16]	0.14 [0.05, 0.21]
Lemmer & Wagner 2015	123	0.73 [0.58, 0.89]	not flagged	–0.13 [–0.26, –0.02]	0.33 [0.27, 0.38]	0.39 [0.31, 0.48]
Burnes et al. 2019	57	0.71 [0.49, 0.88]	not flagged	0.15 [–0.01, 0.33]	0.35 [0.18, 0.53]	–0.36 [–1.38, 0.47]
da Costa et al. 2024	91	0.67 [0.46, 0.86]	flagged	0.21 [0.05, 0.35]	0.24 [0.13, 0.36]	0.12 [–0.27, 0.35]
Maunder & White 2019	109	0.66 [0.47, 0.86]	flagged	0.13 [–0.07, 0.26]	0.41 [0.33, 0.53]	0.45 [0.28, 0.65]
McManus et al. 2021	54	0.63 [0.40, 0.84]	flagged	0.21 [–0.12, 0.45]	0.71 [0.50, 0.91]	0.62 [0.23, 1.03]
Hsieh et al. 2022	39	0.61 [0.40, 0.80]	flagged	0.29 [0.11, 0.48]	0.22 [–0.04, 0.46]	–0.17 [–0.62, 0.33]

Point [95% bootstrap CI]; full samples, no cross-meta deduplication; ordered by ERR within group. Means are on each group's scale and cannot be compared across groups. Excess significance is flagged when TESS or PSST exceeds 1.645.

† Fewer than ~15 significant results: z-curve unstable, ERR not to be read.

‡ Valenced source: the Q2/Q3 cells show the valence-split values (+ positive, – negative contact, each oriented to its own hypothesis).

The three questions across the 21 meta-analyses

Meta-analysis	k	Q1: ERR	Q2: excess significance	Q2: PSSS	Q3: naive mean	Q3: 3PSM-adjusted
Tondok et al. 2024	18	0.58 [0.31, 0.79]	not flagged	0.47 [–0.41, 1.00]	0.32 [0.25, 0.39]	0.24 [–0.79, 0.39]
Paluck et al. 2021	155	0.41 [0.24, 0.62]	flagged	0.19 [0.09, 0.31]	0.36 [0.28, 0.43]	0.28 [0.16, 0.42]
Miles & Crisp 2014	71	0.33 [0.14, 0.55]	flagged	0.22 [0.11, 0.36]	0.36 [0.25, 0.45]	0.18 [0.01, 0.34]
Griffiths et al. 2014†	5	—	not flagged	0.05 [–0.21, 0.55]	0.33 [0.12, 0.52]	0.29 [0.07, 0.55]
Armstrong et al. 2017†	9	—	flagged	0.35 [–0.09, 0.78]	0.44 [0.18, 0.75]	0.24 [0.12, 0.66]
Rahmani et al. 2025†	22	—	not flagged	–0.13 [–0.47, 0.16]	0.42 [0.14, 0.84]	0.68 [0.12, 1.38]
Collective action (replicability only)						

Point [95% bootstrap CI]; full samples, no cross-meta deduplication; ordered by ERR within group. Means are on each group's scale and cannot be compared across groups. Excess significance is flagged when TESS or PSST exceeds 1.645.

† Fewer than ~15 significant results: z-curve unstable, ERR not to be read.

‡ Valenced source: the Q2/Q3 cells show the valence-split values (+ positive, – negative contact, each oriented to its own hypothesis).

The three questions across the 21 meta-analyses

Meta-analysis	k	Q1: ERR	Q2: excess significance	Q2: PSSS	Q3: naive mean	Q3: 3PSM-adjusted
Reimer & Sengupta 2023	114	0.69 [0.53, 0.85]	—	—	—	—

Point [95% bootstrap CI]; full samples, no cross-meta deduplication; ordered by ERR within group. Means are on each group's scale and cannot be compared across groups. Excess significance is flagged when TESS or PSST exceeds 1.645.

† Fewer than ~15 significant results: z-curve unstable, ERR not to be read.

‡ Valenced source: the Q2/Q3 cells show the valence-split values (+ positive, – negative contact, each oriented to its own hypothesis).

10 Appendix: how each focal test was built

Each meta-analysis reports effects in its own scale and at its own level of nesting. The table below records, for the eight sources with purpose-built focal-test constructions (used by the case studies and subgroup contrasts), the focal test the engine draws (effect size and its standard error), the unit it resamples over to respect dependence, and how the sign was oriented so that positive means the hypothesised prejudice reduction. Full provenance, filtering rules and validation checks are in the separate Data Extraction Appendix (`extraction_appendix.qmd`).

Meta-analysis	Focal test (effect size / SE)	Resampling unit	Orientation
Pettigrew & Tropp (2006)	r, SE = $1/\sqrt{(n-3)}$	sample (714, one per row)	sign flipped: negative r = less prejudice
Miles & Crisp (2014)	Cohen's d, two-cell designs	experiment (one per row)	positive d = better attitudes
Paluck et al. (2021)	authors' d / SE	study (155; 508 effects)	authors' d already oriented

Meta-analysis	Focal test (effect size / SE)	Resampling unit	Orientation
Zhou et al. (2019)	*r*, SE = 1/√(n−3)	study (95; 182 effects)	valence-oriented (Negative-valence rows flipped)
da Costa et al. (2024)	Hedges's *g* = yi/√vi	sample: Paper × Study × Sample_ID (91; 293 effects)	sign flipped: negative *g* = less prejudice
Hsieh et al. (2022)	*d* / √v, contact arm	study (40; 55 arms)	positive *d* = less prejudice
Lemmer & Wagner (2015)	immediate *g*, design-appropriate SE	comparison (123, one per row)	positive *g* = less prejudice
Lowe (2025)	preregistered effect / SE / N	experiment (25; 113 effects)	positive effect = intended direction

The field-wide sections quote each meta-analysis's full contact sample; the case studies and subgroup contrasts quote these focal constructions. For most sources the two coincide or differ trivially: Pettigrew & Tropp, Paluck and Lemmer are identical, and Hsieh and da Costa differ by a handful of effects with essentially unchanged estimates. Two differences matter. Zhou's focal set is extended contact only (ERR 0.82 vs 0.87 on the full sample, which adds cross-group friendship), and only the focal set flags selection (Section 4). Lowe's 25-experiment focal set has too few significant results for a stable z-curve, whereas the full 41-effect sample is estimable (Section 6.4); its mean-effect estimates are similar on both, though some full-sample CIs exclude zero.

Three of these constructions involve approximations that carry into the results and are flagged again where they bite:

- **Lowe (2025)** is used at the effect level: the replication package carries a matched standardised effect, SE and sample size for every clean-contact comparison (113 effects across the same 25 experiments as the paper's own "Clean" specification), so the full battery, including ML power, runs, with the engine's one-effect-per-experiment resampling standing in for Lowe's own aggregation step.
- **Lemmer & Wagner (2015)** report one *g* and one *N* per comparison; the test z is reconstructed with a design-appropriate standard error (between-group vs within-group PPSG), so its inputs are approximate.
- **Zhou et al. (2019)** codes each *r* in its outcome's native valence: for "Negative"-valence outcomes (prejudice-type measures) the hypothesised effect is a *negative* r. Effects are

therefore valence-oriented before analysis, and the oriented random-effects mean (Fisher z 0.26, r ≈ .25) reproduces the published pooled r = .25, which validates the orientation.

The other thirteen meta-analyses in the pooled database are constructed the same way — one focal effect size and standard error per contact effect, a participant-pool resampling unit, and a sign oriented so that positive is prejudice/stigma reduction — in `build_pooled_db.R`, which harmonises all 21 sources to a common schema. Two orientations were flagged there and confirmed against the published pooled estimate before any direction-dependent statistic: **Paolini et al. (2024)** codes *r* in Pettigrew & Tropp style (positive = *more* prejudice), so its sign is flipped; and **da Costa et al. (2024)** codes prejudice reduction as a negative *g*, also flipped. For every source the inverse-variance-weighted oriented mean was checked to be positive and, where a published pooled estimate exists, to match it in sign and rough magnitude (for example Pettigrew & Tropp r ≈ .21, Zhou r ≈ .25, McManus *d* ≈ .44, Burnes *d* ≈ .33, Tondok *g* ≈ .32). Two constructions carry approximations noted in Section 12: Burnes's effect sizes are read from forest-plot figures, and Apriceno & Levy report no per-study *N*, so their standard errors come from the printed confidence intervals (which also disables ML power for that meta).

10.1 Subgroup contrasts within the focal-test sets

dataset	subgroup	ERR	PSSS
Pettigrew & Tropp 2006	design=1 survey/field	0.76 [0.69, 0.82]	−0.04 [−0.17, 0.07]
Pettigrew & Tropp 2006	design=2 quasi-exp	0.60 [0.47, 0.72]	0.02 [−0.12, 0.13]
Pettigrew & Tropp 2006	design=3 experiment	0.51 [0.27, 0.77]	−0.14 [−0.57, 0.16]
Pettigrew & Tropp 2006	published=published	0.69 [0.63, 0.77]	−0.00 [−0.11, 0.09]
Pettigrew & Tropp 2006	published=unpublished	0.75 [0.60, 0.86]	−0.06 [−0.27, 0.11]
Paluck et al. 2021	approach=direct	0.48 [0.03, 0.80]	0.16 [−0.01, 0.38]
Paluck et al. 2021	approach=indirect	0.41 [0.23, 0.59]	0.20 [0.09, 0.33]
Paluck et al. 2021	setting=field	—	0.22 [−0.09, 0.63]
Paluck et al. 2021	setting=lab	0.41 [0.22, 0.58]	0.18 [0.05, 0.31]
Paluck et al. 2021	setting=online	0.50 [0.15, 0.82]	0.03 [−0.14, 0.22]
da Costa et al. 2024	experimental=correlational	—	0.43 [−0.11, 1.00]

dataset	subgroup	ERR	PSSS
da Costa et al. 2024	experimental=experimental	0.61 [0.40, 0.81]	0.16 [0.01, 0.31]
Lemmer & Wagner 2015	direct=direct	0.84 [0.71, 0.96]	−0.12 [−0.27, −0.00]
Lemmer & Wagner 2015	direct=indirect	—	−0.11 [−0.36, 0.14]

dataset	contrast	Δ ERR [95% CI]	verdict
Pettigrew & Tropp 2006	design: 1 survey/field - 2 quasi-exp	0.15 [0.03, 0.30]	differ (CI excludes 0)
Pettigrew & Tropp 2006	design: 1 survey/field - 3 experiment	0.23 [−0.04, 0.49]	n.s.
Pettigrew & Tropp 2006	design: 2 quasi-exp - 3 experiment	0.07 [−0.17, 0.36]	n.s.
Pettigrew & Tropp 2006	published: published - unpublished	−0.04 [−0.18, 0.13]	n.s.
Paluck et al. 2021	approach: direct - indirect	0.04 [−0.39, 0.44]	n.s.
Paluck et al. 2021	setting: field - lab	—	unreliable (an arm has too few significant results)
Paluck et al. 2021	setting: field - online	—	unreliable (an arm has too few significant results)
Paluck et al. 2021	setting: lab - online	−0.02 [−0.55, 0.40]	n.s.
da Costa et al. 2024	experimental: correlational - experimental	—	unreliable (an arm has too few significant results)
Lemmer & Wagner 2015	direct: direct - indirect	0.60 [0.56, 0.66]	unreliable (an arm has too few significant results)

Only one contrast is both well-estimated and significant: the within-Pettigrew & Tropp design gradient discussed in Section 6.1. The remaining contrasts cannot be read as differences: in Paluck's settings, da Costa's experimental-vs-correlational split, and Lemmer's direct-vs-indirect split, one arm has too few significant results for an ERR to be estimated at all (its bootstrap is nearly empty, which manufactures a spuriously narrow CI). The usable readings within them point the expected way: Paluck's indirect experiments carry more selection than its direct ones (PSSS 0.20 vs 0.16), and Lemmer's direct face-to-face interventions carry the highest ERR among the focal-set subgroup estimates (0.84). The indirect arms, though, are simply too small to compare against.

11 Caveats

- **The p-value construction is approximate.** The z-curve and p-curve inputs are p-values computed from each meta-analysis's effect sizes (r + N via the t-distribution; d or g via effect/SE), not the primary studies' own reported tests; the correspondence is close but not exact, especially for small samples and within-subject designs.
- **The design and contact-type cuts are marginal, not joint.** Neither holds the other constant, so "contact type does not order replicability" is a statement about the marginal pattern; a design × type joint model is needed to show type adds nothing within design (Section 3.1).
- **p-curve and the small-*k* problem.** p-curve is biased upward under heterogeneity, and any literature with fewer than about 15 significant results yields unstable z-curve/p-curve fits and wide or missing CIs, which are flagged in the tables rather than read as precise.
- **da Costa direction.** da Costa's outcome is coded so that a negative value is prejudice reduction (its reported pooled effect is a reduction); the direction-dependent statistics (PSST/TESS, the 3PSM negative-significant weight) assume that orientation.
- **Valenced sources are split for direction-dependent statistics.** Paolini's and Banas's records mix positive- and negative-contact effects, whose hypothesised directions oppose; their pooled means would offset by construction. Their adjusted means and per-valence selection statistics therefore come from valence-split subsets oriented to each hypothesis (Section 4, Section 5); the pooled by-meta rows remain valid for the sign-free power statistics (ERR, EDR, ODR), but their pooled TESS/PSSS treat hypothesis-consistent negative-contact findings as wrong-direction results and understate selection — the split values are the ones to read.
- **Lemmer & Wagner SE is reconstructed** from *g* and *N* with a design-appropriate but approximate standard error.
- **Lowe's case-study z-curve rests on too few significant results to read** (~7 per resample); the rest of its battery runs in full at the effect level, but with 25 independent experiments the selective-reporting and adjusted estimates are imprecise.

12 Data availability of contact meta-analyses

All 21 analysed sources passed the same validation — the oriented effects reproduce the published pooled estimate (Section 10) — but they differ in what kind of per-study data could be obtained, which determines how much reconstruction the focal tests rest on:

Machine-readable datasets (authors' repository, replication package or supplementary data file): Pettigrew & Tropp (2006, SPSS digitization of the paper's study-level appendix), Zhou et al. (2019, OSF `4fygb`), da Costa et al. (2024, OSF `xjqzn`), Paluck et al. (2021, Harvard Dataverse `ODACR5`), Hsieh et al. (2022, OSF `d7k93`), Lowe (2025, author's replication package: per-comparison standardised effects, SEs and sample sizes), Paolini et al. (2024, OSF Appendix F3 workbook), Reimer & Sengupta (2023, authors' GitHub/OSF data), and Rahmani et al. (2025, authors' data workbook).

Tables re-keyed from the paper or its supplement: Miles & Crisp (2014, in-paper Table 1), Lemmer & Wagner (2015, Wiley Supporting Information C), Smith et al. (2009, in-paper Table 1), Banas et al. (2020, in-paper Table 1), Maunder & White (2019, Appendix B), Griffiths et al. (2014, in-paper tables), Armstrong et al. (2017), McManus et al. (2021, in-paper tables), Van Assche et al. (2023, supplementary correlation tables), and Apriceno & Levy (2023) (Hedges's g with confidence intervals but no per-study N, so its standard errors are reconstructed from the intervals).

Digitized from figures: Burnes et al. (2019) (effect sizes read off forest-plot figures, joined to the tabled N and contact codes) and Tondok et al. (2024) (g with confidence intervals from the article's study table and forest plot).

No per-study data available (only pooled/moderator estimates published, so still un-auditable): Davies et al. (2011) on cross-group friendship and Beelmann & Heinemann (2014).

13 Reproducibility

The engine and its run are in `replicability_engine/`: `engine.R` (`run_battery()` — the reusable battery), `run_contact_battery.R` (builds the eight focal-test tables and runs it), `run_two_batteries.R` (the by-meta and by-stratum runs on the pooled database), `run_valence_splits.R` (the per-hypothesis valence subsets of Paolini and Banas), `rr_helpers/` (the registered report's verbatim statistical helpers), `es_adapt.R` (the adapted effect-size-based ESS/3PSM and the bias-adjusted estimators), and the outputs `battery_results.csv`, `by_meta_results.csv`, `by_stratum_results.csv`, `by_valence_results.csv`, `battery_summary.txt`, `fig_z_*.png`. The pooled database is built by `build_pooled_db.R`. See `replicability_engine/README.md` for run time, iteration counts and per-dataset caveats. The SCORE-based validation of ARP is in `zcurve_predictive_accuracy/` (data provenance in its README, results in `FINDINGS.md`); the per-dataset extraction documentation is in `extraction_appendix.qmd`.

The engine flags two routine conditions across most datasets that are not signs of a problem: ML power is reported as a point estimate with its bootstrap CI omitted (Section 8.4), and the 3PSM falls back to a one-sided selection model wherever a dataset has too few negative-significant results to fit the negative-weight term. The only substantive instability flag in the focal-set battery is **Lowe (2025)**, whose z-curve rests on about seven significant results per resample (86–100% of its bootstrap fits return NA); its ERR is reported but not interpreted (Section 6.4).

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